Induced Subgraphs in Ramsey Graphs

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Abstract

In this dissertation, we investigate the "random-like" features of Ramsey graphs, with an emphasis on the existence of induced subgraphs of many different sizes.

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1 Introduction

Classical Ramsey Theory asks for the minimum number of vertices required to guarantee the existence of a clique or an independent set in a graph. We refer to these maxima as *Ramsey numbers*.

Definition 1.1. The *Ramsey number* $R(s,t)$ is the minimum number N such that any graph on N vertices contains a clique of size s or an independent set of size t.

The existence of these numbers was proven by Ramsey [15] in 1930, but their quantitative behaviors remain largely unknown despite decades of intensive research. In particular, it is known that $R(3,3) = 6$ and $R(4,4) = 18$, but the exact values of $R(5,5)$ and beyond are yet to be determined. The cornerstone of Ramsey Theory due to Erdős and Szekeres in 1935 gives the following upper bound on $R(s,t)$:

Theorem 1.1 (Erdős–Szekeres [8])**.** *For $s,t \geq 2$,*

$$R(s,t) \leq \binom{s+t-2}{s-1}.$$

Following the fact that $\binom{2t-2}{t-1} < 2^{2t}$, we can see $R(t,t) < 4^t$. Very little progress had been made on this bound until the recent breakthrough of Campos, Griffiths, Morris, and Sahasrabudhe [4], who gave the first exponential improvement since the result of Erdős and Szekeres. On the other hand, the lower bound of ramsey numbers has been even more challenging to determine. A rather basic result which already gives the best known exponents is the famous random graph construction due to Erdős in 1947, marking one of the first applications of the probabilistic method in combinatorics. Before stating the result, we first define the *Erdős–Rényi random graph*.

Definition 1.2. For $n \geq 1$ and $p \in [0,1]$, the *random graph* $G_{n,p}$ is the graph on n vertices in which each of the $\binom{n}{2}$ edges is present independently with probability p.

Remark: For constant p, the random graph $G_{n,p}$ is dense in the sense that it has $\Theta(n^2)$ many edges almost surely.

Theorem 1.2 (Erdős [9])**.** *For $t \geq 3$,*

$$\sqrt{2}^t < R(t,t) < 4^t.$$

Proof. The upper bound follows from Theorem 1.1. For lower bound, set $N = \lfloor \sqrt{2}^t \rfloor$ and consider the random graph $G_{N,1/2}$. The expected number of cliques of size t is

$$\binom{N}{t} \cdot 2^{-\binom{t}{2}} < \frac{N^t}{t!} 2^{-\binom{t}{2}} \leq \frac{\sqrt{2}^{t^2}}{t!} 2^{-\binom{t}{2}} = \frac{\sqrt{2}^t}{t!} = \frac{\sqrt{2}^{t-3}}{t(t-1)\cdots 4} \cdot \frac{\sqrt{2}^3}{3!} \leq \frac{\sqrt{2}^3}{3!} < \frac{1}{2}.$$

Similarly, the expected number of independent sets of size t is less than $1/2$. It now follows that there exists a graph on N vertices containing neither a clique nor an independent set of size t. □

The above strategy may be optimized to give a lower bound of $\frac{t}{e\sqrt{2}}\sqrt{2}^t$, which has been astonishingly resilient to improvement. The only noteworthy improvement since 1947 is achieved in 1977 by Spencer [17], who used the Lovász local lemma to raise the bound to $(1-o(1))\frac{t\sqrt{2}}{e}\sqrt{2}^t$, an improvement by a factor of 2. For further background on Ramsey theory, see Conlon, Fox, and Sudakov [6].

From Theorem 1.2, we can see that any graph on n vertices contains a clique or independent set of size at least $\frac{1}{2}\log_2 n$, whereas there exists a graph on n vertices containing neither a clique nor an independent set of size $2\log_2 n$. Apart from determining the bounds of Ramsey numbers, the search for explicit graph constructions with logarithmically small cliques and independent sets has also been a major open problem in Ramsey Theory. Despite decades of intensive research, all known constructions still rely on some randomness. This thus motivates the question of whether graphs with small cliques and independent sets inherently resemble dense random graphs like $G_{n,1/2}$; we refer to such graphs as *Ramsey graphs*.

Definition 1.3. Let $C > 0$ be a constant. A *C-Ramsey graph* is a graph with no clique or independent set of size greater than $C\log n$.

It is useful to also define a shorthand for cliques and independent sets.

Definition 1.4. A set $S \subseteq V(G)$ is *homogeneous* in a graph G if it induces a clique or an independent set in G. We denote the size of the largest homogeneous set in G by $\hom(G)$.

The belief that Ramsey graphs and random graphs are intimately related is supported a number of theorems showing that certain "richness" properties of random graphs also hold for Ramsey graphs. The first result that cemented this connection is due to Erdős and Szemerédi, showing that Ramsey graphs are also dense.

Theorem 1.3 (Erdős–Szemerédi [9])**.** *Let G be a graph on n vertices with edge density $d \le 1/2$. Then G contains a homogeneous set of size greater than $K = \frac{c}{d\log(1/d)}\log n$, for some constant $c > 0$ and sufficiently large n.*

Proof. Suppose not. Pick $c < 1/32$. Since G contains $d\binom{n}{2}$ edges, there exists $m \ge n/2$ vertices in G with degree at most dn, by the pigeonhole principle. We work with the graph G' induced by these m vertices. Let I be the largest independent set in G' and let $k = |I|$. Since $\deg(v) \le dn$ for all $v \in I$, we have $\sum_{v\in I}\deg(v) \le dnk \le 2dmk$. Put $s = 8dk$. Again by the pigeonhole principle, there are at least $m/2 - k$ vertices $v \in V(G')\setminus I$ with $\deg_I(v) < s$. Let J be the set of these vertices. Consider the family $\mathcal{F}$ of subsets of I with size less than s. Partition J by each vertex's neighbourhood in I, and note that each resulting class corresponds to an element of $\mathcal{F}$. By the standard bounds $\binom{x}{r} \le (ex/r)^r$ and $1 + x \le e^x$,

$$|\mathcal{F}| \le s\binom{k}{s-1} \le s\left(\frac{ek}{s-1}\right)^{s-1} = s\left(\frac{s}{s-1}\right)^{s-1}\left(\frac{ek}{s}\right)^{s-1} \le se\left(\frac{ek}{s}\right)^{s-1} = \frac{s^2}{k}\left(\frac{ek}{s}\right)^{s}.$$

By assumption $|I| = k < K$, and so

$$s < 8dK = \frac{8c\log n}{\log(1/d)} = O(\log n).$$

Since $s/k = 8d$, it follows that

$$|\mathcal{F}| \leq \frac{s^2}{k}\left(\frac{ek}{s}\right)^s \leq O(\log n)\cdot\left(\frac{e}{8}\cdot\frac{1}{d}\right)^s \leq O(\log n)\cdot\left(\frac{1}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = O(\log n)\cdot n^{8c} < n^{1/3}.$$

But then

$$\frac{|J|}{|\mathcal{F}|} \geq \frac{m/2-k}{n^{1/3}} \geq \sqrt{n},$$

so by the pigeonhole principle, there exists a class of vertices Z in J of size at least $\sqrt{n}$. Let $G'' = G[Z]$. Note that G'' does not contain an independent set of size s, otherwise exchange it with Y in I creates a larger independent set in G', contradicting the maximality of I. Let t be the size of the largest clique in G'', and note that $t < K$ by assumption. The Erdős-Szekeres and the standard binomial bounds give

$$\sqrt{n} \leq |Z| \leq \binom{s+K-2}{s-1} \leq \binom{s+K}{s} \leq \binom{8dK+K}{8dK} \leq \left(\frac{e(8d+1)}{8d}\right)^s \leq \left(\frac{2}{d}\right)^s,$$

where the last inequality holds because $e(1+8d)/8 \leq 5e/8 \leq 2$ for $d \leq 1/2$. But then since $d \leq 1/2$ and $c < 1/32$,

$$\left(\frac{2}{d}\right)^s \leq \left(\frac{2}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = n^{8c\cdot\frac{\log(2/d)}{\log(1/d)}} = n^{8c\left(1+\frac{\log 2}{\log(1/d)}\right)} \leq n^{16c} < \sqrt{n},$$

contradiction. □

Recall that the Erdős–Szemerédi theorem gives the bound $\hom(G) \geq \frac{1}{2}\log_2 n$, regardless of the edge density. Theorem 1.3 improves this when the density is small. For example, if $d = 1/n^\varepsilon$ for some $\varepsilon > 0$, then $\hom(G) = \Omega\left(\frac{n^\varepsilon}{\log n^\varepsilon}\log n\right)$, vastly larger than $\frac{1}{2}\log_2 n$. Intuitively, a sparse graph should contain a large independent set, and the theorem makes this notion precise.

If G is C-Ramsey, then applying Theorem 1.3 to whichever G or $\overline{G}$ has density $d \leq 1/2$ yields $c/(d\log(1/d)) \leq C$, forcing d to be bounded away from 0. By the same argument applied to $\overline{G}$ forces d to be bounded away from 1. We thus have the following corollary.

Corollary 1.4. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Due to the rich variety of induced subgraphs in dense random graphs, a popular angle to study the pseudorandom properties of Ramsey graphs is to consider their induced subgraphs. Some notable results are a theorem of Shelah [16] which asserts

that every n-vertex Ramsey graph contains $2^{\Omega(n)}$ non-isomorphic induced subgraphs, and a theorem of Prömel and Rödl [14] which shows the existence of $\delta = \delta(C) > 0$ such that every C-Ramsey graph on n vertices contains an induced copy of all graphs on at most $\delta \log n$ vertices.

It is also interesting to look at the edge statistics of Ramsey graphs. For a graph G, let

$$\Phi(G) \coloneqq \{e(H) : H \text{ is an induced subgraph of } G\}$$

Erdős and McKay [7] conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G).$$

It was proven for Random graphs by Calkin, Frieze, and Mckay [3], but the conjecture for Ramsey graphs remained open for decades and only started to show significant progress in recent years. In particular, a chain of results were sparked after the work of Narayanan, Sahasrabudhe, and Tomon [13] establishing that $|\Phi(G)| = n^{2-o(1)}$ for Ramsey graph G on n vertices. Their proof builds on some ideas from the work of Bukh and Sudakov [2] which showed that every Ramsey graph contains a linear sized subgraph with many distinct degrees, a feature that dense random graphs also exhibit. Kwan and Sudakov [11] later improved the bound to $|\Phi(G)| = \Omega(n^2)$ by refining the argument of Narayanan, Sahasrabudhe, and Tomon. Eventually, Kwan, Sah, Sauermann, and Sawhney [12] closed the conjecture with a fairly involved approach using tools from Fourier analysis, random matrix theory, and the theory of Boolean functions.

This dissertation will start by studying the result of Bukh and Sudakov [2] on large induced subgraphs in Ramsey graphs with many distinct degrees. This work not only tightens the connection between Ramsey graphs and random graphs, but also introduces a key feature of Ramsey graphs that unlocks the subsequent $|\Phi(G)| = \Omega(n^2)$ result by Kwan and Sudakov, which we will present in the following chapter. Lastly, we go through the broad strokes of the proof to the Erdős–McKay conjecture by Kwan, Sah, Sauermann, and Sawhney [12], touching on the core ideas of their approach.

1.1 Notations and Tools

We call $e(G)$ the *size* of a graph G. We omit all floor and ceiling signs when they are not crucial for simplicity.

Proposition 1.5 (Turán's theorem)**.** *Every graph G with n vertices and m edges contains an independent set of size at least $n^2/(2m+n)$.*

Proposition 1.6 (Hoeffding's inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent real-valued random variables with $0 \le X_i \le 1$ for each i, and write $X = \sum_{i=1}^n X_i$. Then for $t \ge 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \ge t) \le 2e^{-2t^2/n}.$$

Proposition 1.7. *Let X denote the number of successes in D draws, without replacement, from a population of size N which contains M successes. Then $\mathbb{E}[X] = DM/N$ and for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2 D}.$$

Proof. □

1.2 The Anticoncentration Principle

Before diving into the core results, we introduce a key probabilistic mechanism which serves as a unifying thread throughout the dissertation. In the following chapters, we will repeatedly sample a random vertex set $U \subseteq V(G)$ and study the induced subgraph $G[U]$. One natural question to ask in this setting is how likely are two vertices x and y to have the same degree in $G[U]$?

Observe that $\deg_U(x) - \deg_U(y)$ counts only vertices in the symmetric difference $D = N_G(x) \Delta N_G(y)$. Writing $A = N(x) \setminus N(y)$ and $B = N(y) \setminus N(x)$, we have

$$\deg_U(x) - \deg_U(y) = |U \cap A| - |U \cap B|, \tag{1}$$

x and y have the same degree in $G[U]$ precisely when $|U \cap A| = |U \cap B|$.

Suppose first that U is formed by including each vertex independently with probability p. Then each vertex $v \in A$ independently contributes $+1$ to the difference $|U \cap A| - |U \cap B|$ with probability p, while each vertex $v \in B$ independently contributes -1 to the difference with probability p. The difference is thus a sum of $|D| = |A| + |B|$ independent bounded random variables. When $|D|$ is large, say $|D| \geq \varepsilon n$, this sum has standard deviation $\Theta(\sqrt{n})$. Since $|U \cap A| - |U \cap B|$ is integer-valued, we can show the probability that it equals any particular integer is at most $O(1/\sqrt{n})$ via the one-dimensional Erdős-Littlewood-Offord theorem. Thus we conclude that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$. We will formalize this with greater detail in Chapter 3.

If instead U is a uniformly random subset of a fixed size m, the vertices of U are no longer independent. Nevertheless, conditioning on $k = |U \cap D|$, we may observe that, given k, the split of these k elements between A and B follows a hypergeometric distribution. With the same assumption that $|D|$ is large, Stirling's approximation applied to the hypergeometric probability then yields the same $O(1/\sqrt{n})$ bound. We will carry out this calculation in detail in Chapter 2.

An intuition to the bound $O(1/\sqrt{n})$ is that, in either sampling method, the difference $\deg_U(x) - \deg_U(y)$ is approximately normally distributed with standard deviation $\Theta(\sqrt{n})$. In the independent case, this follows from the central limit theorem applied to a sum of independent bounded random variables. In the fixed-size case, it follows from the normal approximation to the hypergeometric distribution. Since the probability density function $\phi(x)$ of a normal distribution with standard deviation σ is at most $1/(\sigma\sqrt{2\pi})$ for all x, the probability that $\deg_U(x) - \deg_U(y)$ equals any particular

value is $O(1/\sigma) = O(1/\sqrt{n})$. The theorems in the following chapters will rely on this principle. In Chapter 2, it bounds the probability that two vertices share a degree in a random induced subgraph, which leads to $\Omega(\sqrt{n})$ distinct degrees. In Chapter 3, the same principle bounds the probability that two vertices have equal degree in a random induced subgraph, which is a key step in producing $\Omega(n^2)$ distinct subgraph sizes. The resolution of the Erdős–McKay conjecture in Chapter 4 requires a substantially more powerful approach via Fourier analysis, but the underlying idea remains: when a random variable is approximately normal with large standard deviation, its point probabilities are small.

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [2])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [7] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely, but the $\Omega(\sqrt{n})$ bound is in fact not optimal for both Ramsey graphs and $G_{n,1/2}$. In particular, an upper bound of $O(n^{2/3})$ is noted in [2], which was later proved to be the correct order for Ramsey graphs by Jenssen, Keevash, Long, and Yepremyan [10] via a continuous relaxation approach. The corresponding order for $G_{n,1/2}$ was also matched in an unpublished work of Conlon, Morris, Samotij, and Saxton [5], further reinforcing the case that Ramsey graphs resemble random graphs in various respects. We will, however, focus on the original result of Bukh and Sudakov, as it introduces a key characteristic of Ramsey graphs that will be insightful in the following chapter.

Definition 2.1. An n-vertex graph G is (ε, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| \geq \varepsilon n$ for all but at most n^δ vertices $y \in V(G)$.

The notion of diversity captures the idea that the neighbourhoods of most pairs of vertices differ significantly. In particular, (ε, δ)-diversity directly conditions our assumption that $|N(x)\Delta N(y)| = D \geq \varepsilon n$ in the discussion of the anticoncentration principle in Section 1.2. Hence, following our discussion in Section 1.2, we see that (ε, δ)-diversity implies that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$ for all but at most n^δ pairs of vertices $x, y \in V(G)$, from which we use Jensen's inequality to conclude that there are many distinct degrees in a diverse graph. The following lemma formalizes this observation and reduces Theorem 2.1 to showing every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.2. *Let $\delta \in (0, 1/2)$ and let G be an (ε, δ)-diverse graph on n vertices for large enough n. Then for $n/4 \leq m \leq 3n/4$, there is a subgraph on m vertices containing vertices of $\sqrt{\varepsilon n}/320$ distinct degrees.*

Proof. Fix m as above. Let U be chosen uniformly at random among all subsets of $V(G)$ of size m. Let $H = G[U]$. It suffices to show that with positive probability, H contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in V(H)$ and $\deg_H(x) = \deg_H(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \leq \mathbb{P}[\deg_U(x) = \deg_U(y)]$. Put $D = N_G(x)\Delta N_G(y)$, and $k = |D \cap U|$. Write $A = N_G(x) \setminus N_G(y)$ and $B = N_G(y) \setminus N_G(x)$. Then

$$\mathbb{P}[\deg_U(x) = \deg_U(y)] = \mathbb{P}\left[|A \cap U| = |B \cap U|\right].$$

Suppose $|D| \geq \varepsilon n$. Since $\frac{|D|}{4} \leq \frac{m|D|}{n} \leq \frac{3|D|}{4}$, Proposition 1.7 yields

$$\mathbb{P}\left[k < \frac{|D|}{8} \text{ or } k > \frac{7|D|}{8}\right] \leq \mathbb{P}\left[|k - \mathbb{E}[k]| > \frac{|D|}{8}\right] \leq e^{-\Theta(|D|)} = e^{-\Theta(n)} = o(n^{-1/2}).$$

On the other hand, recall that U is a uniformly random subset of a fixed size m. Since every k-subset of D is equally likely to be $U \cap D$ by symmetry, conditioning on $|U \cap D| = k$, the k elements of U that land in D form a uniformly random subset of D of size k. It follows that when conditioning on k, the event $|A \cap U| = |B \cap U| = k/2$ follows the hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$. Thus,

$$\mathbb{P}\left[\deg_U(x) = \deg_U(y) \,\middle|\, \frac{|D|}{8} \leq k \leq \frac{7|D|}{8}\right] = \frac{\binom{|A|}{k/2}\binom{|B|}{k/2}}{\binom{|D|}{k}}.$$

Since $|A| + |B| = |D|$, by the log-concavity of the binomial coefficients we have $\binom{|A|}{k/2}\binom{|B|}{k/2} \leq \binom{|D|/2}{k/2}^2$. It now follows from Stirling's formula and $k(|D| - k) \geq \frac{7}{64}|D|^2$ that the above probability does not exceed

$$\frac{\binom{|D|/2}{k/2}^2}{\binom{|D|}{k}} = (1 + o(1))\sqrt{\frac{2|D|}{\pi k(|D| - k)}} \leq \frac{3}{\sqrt{|D|}} \leq \frac{3}{\sqrt{\varepsilon n}}. \tag{2}$$

Thus if $|N_G(x) \Delta N_G(y)| \geq \varepsilon n$ then $\mathbb{E}[\mathbb{1}_{xy}] \leq 3/\sqrt{\varepsilon n} + o(n^{-1/2}) \leq 4/\sqrt{\varepsilon n}$.

Let $I = \sum_{x,y \in V(G)} \mathbb{1}_{xy}$. By definition of an (ε, δ)-diverse graph, we have $\mathbb{E}[\mathbb{1}_{xy}] \leq 4/\sqrt{\varepsilon n}$ for all but at most $n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Thus,

$$\mathbb{E}[I] \leq n^{1+\delta} + \frac{4}{\sqrt{\varepsilon n}} \cdot n^2 \leq \frac{5}{\sqrt{\varepsilon}} n^{3/2},$$

and so there is a choice for U such that $I \leq 5n^{3/2}/\sqrt{\varepsilon}$. Now let r be the number of vertices of distinct degrees in H. Let n_k be the number of vertices in H with degree k and note that $\sum_k n_k = m$. By Jensen's inequality

$$5n^{3/2}/\sqrt{\varepsilon} \geq I = \sum_k \binom{n_k}{2} \geq r\binom{\frac{1}{r}\sum_k n_k}{2} \geq \frac{m^2}{4r} \geq \frac{n^2}{64r}.$$

Rearranging yields the lemma. □

Remark. The hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$ has variance

$$\sigma^2 = \frac{k|A||B|(|D| - k)}{|D|^2(D - 1)} \leq \frac{k(|D| - k)}{4(|D| - 1)},$$

where inequality follows from AM-GM. When k is not the tail end of the distribution, we have $\sigma = \Theta(\sqrt{D}) = \Theta(\sqrt{n})$, and the Stirling bound in (2) is asymptotically $1/(\sigma\sqrt{2\pi})$ as $n \to \infty$, which is exactly the maximum density of a normal distribution

with standard deviation σ. This confirms the intuition from Section 1.2 that the degree difference is approximately normally distributed so its point probabilities are $O(1/\sigma) = O(1/\sqrt{n})$.

As mentioned above, it remains to show that every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.3. *For $\delta \in (0,1)$, there exists $\varepsilon = \varepsilon(C,\delta) > 0$ such that every C-Ramsey graph G on n vertices contains an (ε,δ)-diverse subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = K(C)$ be a large constant to be determined. Let $c = 8^{-K}$ and $\varepsilon = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size $(cn)^\delta$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 8\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 8\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \supset G_1 \supset \cdots \supset G_K$, such that $|G_i| \geq |G_{i-1}|/8$ and $S_i \subseteq V(G_{i-1})$ for all i.

We first show that this construction suffices. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size $(cn)^\delta$ and $\varepsilon = 1/K$,

$$d(S) \leq \frac{\binom{(cn)^\delta}{2}}{\binom{r(cn)^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 8\varepsilon \leq \frac{10}{K}.$$

Thus Theorem 1.3 yields a homogeneous set A in S of size

$$|A| \geq \frac{\gamma K}{10\log(K/10)} \log|S| = \frac{\gamma\delta K}{10\log(K/10)} \log cn = \frac{\gamma\delta K}{10\log(K/10)} \log n + o(\log n),$$

for some absolute constant $\gamma > 0$ given by the theorem. But then $\frac{\gamma\delta K}{10\log(K/10)} > C$ for sufficiently large K, and so $|A| > C\log n$, contradiction.

We now construct the sets S_i and G_i inductively. Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|G_{i-1}| \geq 8^{1-i}|G| \geq cn$. Note that G_{i-1} is not (ε,δ)-diverse by assumption, so there exists a vertex $x \in V(G_{i-1})$ and a set $S_i \subseteq G_{i-1}$ of size $(cn)^\delta$ such that $|N_{G_{i-1}}(x) \Delta N_{G_{i-1}}(y)| < \varepsilon|G_{i-1}|$ for $y \in S_i$. Let $T_i = V(G_{i-1}) \setminus S_i$. Since $(cn)^\delta \leq cn/2$ for large enough n, we have $|T_i| \geq |G_{i-1}|/2$.

We consider two cases. In the first case, suppose $|\overline{N_{T_i}}(x)| \geq |T_i|/2$. For $y \in S_i$,

$$|N_{T_i}(y) \cap \overline{N}_{T_i}(x)| \leq |N_{T_i}(x) \Delta N_{T_i}(y)| < \varepsilon|V(G_{i-1})|,$$

and so

$$d(y, \overline{N}_{T_i}(x)) = \frac{|N_{T_i}(y) \cap \overline{N}_{T_i}(x)|}{|N_{T_i}(x)|} < \frac{\varepsilon|V(G_{i-1})|}{|T_i|/2} \leq 4\varepsilon.$$

Partition $\overline{N}_{T_i}(x)$ into $V_i \sqcup Z_i$, where $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) < 8\varepsilon\}$. We double count $e(\overline{N}_{T_i}(x), S_i)/|S_i|$. Since $d(v, S_i) \geq 8\varepsilon$ for $v \in Z_i$,

$$\frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \sum_{v \in \overline{N}_{T_i}(x)} \frac{e(v, S_i)}{|S_i|} = \sum_{v \in \overline{N}_{T_i}(x)} d(v, S_i) \geq \sum_{v \in Z_i} d(v, S_i) \geq 8\varepsilon |Z_i|.$$

On the other hand, since we showed $d(y, \overline{N}_{T_i}(x)) < 4\varepsilon$ for $y \in S_i$,

$$\frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \frac{1}{|S_i|} \sum_{y \in S_i} e(y, \overline{N}_{T_i}(x)) = \frac{|\overline{N}_{T_i}(x)|}{|S_i|} \sum_{y \in S_i} d(y, \overline{N}_{T_i}(x)) \leq 4\varepsilon |\overline{N}_{T_i}(x)|.$$

Combining both inequalities yields $|Z_i| \leq |\overline{N}_{T_i}(x)|/2$. It follows that

$$|V_i| \geq \frac{|\overline{N}_{T_i}(x)|}{2} \geq \frac{|T_i|}{4} \geq \frac{|V(G_{i-1})|}{8}.$$

Recall that $d(S_i, U) < 8\varepsilon$ for all $U \subseteq V_i$. Setting $G_i = G[V_i]$ thus satisfy $|V(G_i)| \geq |V(G_{i-1})|/8$ and condition (i) for S_i.

The case for $|N_{T_i}(x)| \geq |T_i|/2$ follows the same argument. Since any vertex in $N_{T_i}(x)$ that is not a neighbor of y lies in $N_{T_i}(x) \setminus N_{T_i}(y) \subseteq N_{T_i}(x) \Delta N_{T_i}(y)$, we have

$$|N_{T_i}(x) \cap N_{T_i}(y)| \geq |N_{T_i}(x)| - |N_{T_i}(x) \Delta N_{T_i}(y)| < |N_{T_i}(x)| - \varepsilon |V(G_{i-1})|,$$

and so

$$d(y, N_{T_i}(x)) = \frac{|N_{T_i}(y) \cap N_{T_i}(x)|}{|N_{T_i}(x)|} > \frac{|N_{T_i}(x)| - \varepsilon |V(G_{i-1})|}{|N_{T_i}(x)|} \geq 1 - \frac{\varepsilon |V(G_{i-1})|}{|T_i|/2} \geq 1 - 4\varepsilon.$$

Partition $N_{T_i}(x)$ into $V_i \sqcup Z_i$, where $V_i = \{v \in N_{T_i}(x) : d(v, S_i) > 1 - 8\varepsilon\}$. Then for $v \in Z_i$, we have $1 - d(v, S_i) \geq 8\varepsilon$, so

$$8\varepsilon |Z_i| \leq \sum_{v \in Z_i} (1 - d(v, S_i)) \leq \sum_{v \in N_{T_i}(x)} (1 - d(v, S_i)).$$

On the other hand, since we showed $d(y, N_{T_i}(x)) > 1 - 4\varepsilon$ for $y \in S_i$.

$$\frac{e(N_{T_i}(x), S_i)}{|S_i|} = \frac{|N_{T_i}(x)|}{|S_i|} \sum_{y \in S_i} d(y, N_{T_i}(x)) \geq (1 - 4\varepsilon) |N_{T_i}(x)|,$$

But then

$$\sum_{v \in N_{T_i}(x)} (1 - d(v, S_i)) = |N_{T_i}(x)| - \sum_{v \in N_{T_i}(x)} d(v, S_i) \geq 4\varepsilon |N_{T_i}(x)|.$$

Combining both inequalities on $\sum (1 - d(v, S_i))$ yields $|Z_i| \leq |N_{T_i}(x)|/2$, and so $|V_i| \geq\geq |V(G_{i-1})|/8$. Note that $d(S_i, U) > 1 - 8\varepsilon$ for all $U \subseteq V_i$. Setting $G_i = G[V_i]$ thus satisfies $|V(G_i)| \geq |V(G_{i-1})|/8$ and condition (ii) for S_i. This completes the induction. $\square$

Theorem 2.1 follows immediately from Lemmas 2.2 and 2.3.

3 Subgraphs of Quadratically Many Sizes

Recall that Erdős and Mckay conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G) := \{e(H) : H \text{ is an induced subgraph of } G\},$$

That is to say, for any $1 \leq t \leq \delta n^2$, there exists an induced subgraph of size t in G. In [3], the conjecture is proven for random graphs, and thus it is interesting to see if Ramsey graphs also resemble random graphs in this regard. The conjecture was widely open for decades. An early major result was due to Alon, Krivelevich and Sudakov in 2003 [1], where they proved the conjecture with n^δ in place of δn^2, but little progress was made afterwards. Note that Erdős and McKay's conjecture implies that $|\Phi(G)| = \Omega(n^2)$, which is best possible asymptotically as $|\Phi(G)| \leq \binom{n}{2}$. Thus a natural relaxation of the conjecture is to ask if Ramsey graphs really have quadratically many distinct sizes of induced subgraphs. In 2017, Narayanan, Sahasrabudhe, and Tomon [13] proved a near optimal bound of $|\Phi(G)| = n^{2-o(1)}$, whose strategy was built upon the notion of diversity mentioned in Chapter 2. The desired quadratic bound was achieved by Kwan and Sudakov a few years later by refining the strategy of Narayanan, Sahasrabudhe, and Tomon.

Theorem 3.1 (Kwan–Sudakov [11])**.** *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Observe that if a graph G has k distinct degrees, then the induced subgraphs $G - \{v\}$ for $v \in V(G)$ have at least k different sizes. Thus Theorem 2.1 provides a simple bound of $|\Phi(G)| = \Omega(\sqrt{n})$ for any Ramsey graph G. Two straightforward ideas may be used to improve this construction: we can apply our observation to many different induced subgraphs, and, instead of deleting only one vertex, we may add and remove various combinations of vertices to obtain many more distinct sizes.

We will build upon the notion of diversity from Chapter 2. Recall that a graph G is (ε, δ)-diverse if for $x \in V(G)$, we have $|N(x)\Delta N(y)| < \varepsilon n$ for at most n^δ vertices $y \in V(G)$. As in the proof of Theorem 2.1, the significance of diversity is that in a diverse graph, for most pairs of vertices their degrees into some random vertex set U are fairly uncorrelated. In addition, we also introduce the notion of diversity for pairs of vertices. For a pair of vertices $\mathbf{v} = \{v_1, v_2\}$, let $N(\mathbf{v})$ be the multiset union of $N(v_1)$ and $N(v_2)$, and let $d(\mathbf{v}) := |N(\mathbf{v})| = d(v_1) + d(v_2)$.

Definition 3.1. An n-vertex graph G is $(\varepsilon, \delta, \alpha)_2$-*diverse* if for each pair of vertices $\mathbf{x} = \{x_1, x_2\}$ in G with $|N(x_1)\Delta N(x_2)| \geq \alpha n$, there are at most n^δ disjoint pairs of vertices $\mathbf{y} = \{y_1, y_2\}$ with $\mathbf{x} \cap \mathbf{y} = \emptyset$, such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \varepsilon n$.

We introduce a condition slightly stronger than diversity, which would be convenient for our proof.

Definition 3.2. An n-vertex graph G is (ε, δ)-*rich* if, for $W \subseteq V(G)$ with $|W| \geq \delta n$, we have $|N(v) \cap W| \geq \varepsilon |W|$ and $|\overline{N(v)} \cap W| \geq \varepsilon |W|$ for all but at most n^δ vertices

$v \in V(G)$.

Lemma 3.2. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then*

1. *G is $(\varepsilon/2, \delta)$-diverse,*
2. *G is $(\alpha\varepsilon/2, \delta, \alpha)_2$-diverse for any $\alpha \geq 2\delta$,*
3. *G has at most $n^{1+\delta}$ pairs of vertices $x_1, x_2 \in V(G)$ with $|N(x_1)\Delta\overline{N(x_2)}| < (\varepsilon/2)n$.*

Proof. We prove each statement in order.

1. Note that for each $x \in V(G)$, either $|N(x)| \geq n/2$ or $|\overline{N(x)}| \geq n/2$. If $|N(x)| \geq n/2$, then richness implies that
$$|\overline{N(y)} \cap N(x)| \geq \varepsilon|N(x)| \geq \varepsilon n/2$$
for at most n^δ vertices $y \in V(G)$. Similarly, if $|\overline{N(x)}| \geq n/2$, then
$$|N(y) \cap \overline{N(x)}| \geq \varepsilon|\overline{N(x)}| \geq \varepsilon n/2$$
for at most n^δ vertices $y \in V(G)$. But then in either case,
$$|N(x)\Delta N(y)| = |\overline{N(y)} \cap N(x)| + |N(y) \cap \overline{N(x)}| \geq \varepsilon n/2.$$

2. Suppose not. Then there exists a pair $\mathbf{x} = \{x_1, x_2\}$ with $|N(x_1)\Delta\overline{N(x_2)}| \geq \alpha n$ and a set Y of n^δ pairs such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \alpha\varepsilon n/2$ for all $\mathbf{y} \in Y$. Since
$$|N(x_1)\Delta\overline{N(x_2)}| = |N(x_1) \cap N(x_2)| + |\overline{N(x_1)} \cap \overline{N(x_2)}| \geq \alpha n,$$
we have $|N(x_1) \cap N(x_2)| \geq (\alpha/2)n \geq \delta n$ or $|\overline{N(x_1)} \cap \overline{N(x_2)}| \geq (\alpha/2)n \geq \delta n$. Assume without loss of generality that the former case holds. Then for each vertex y in any $\mathbf{y} \in Y$,
$$\left|\overline{N(y)} \cap (N(x_1) \cap N(x_2))\right| \leq |N(\mathbf{x})\Delta N(\mathbf{y})| < \alpha\varepsilon n/2 \leq \varepsilon|N(x_1) \cap N(x_2)|.$$
But then the set of all such y now contradicts the (ε, δ)-rich condition.

3. We show that for each choice of $x_1 \in V(G)$, there are at most n^δ vertices $x_2 \in V(G)$ with $|N(x_1)\Delta\overline{N(x_2)}| < (\varepsilon/2)n$. Since either $|N(x_1)| \geq n/2$ or $|\overline{N(x_1)}| \geq n/2$, we assume without loss of generality that $|N(x_1)| \geq n/2$. By (ε, δ)-richness, there are at most n^δ vertices x_2 with $|N(x_2) \cap N(x_1)| < \varepsilon n/2$. For all other choices of x_2,
$$|N(x_1)\Delta\overline{N(x_2)}| \geq |N(x_1) \cap N(x_2)| \geq (\varepsilon/2)n.$$
The result now follows from the fact that there are only n choices of x_1.

□

We also show the parallel result of Lemma 2.3 that every C-Ramsey graph contains an (ε, δ)-rich subgraph of linear size, following the same line of argument.

Lemma 3.3. *For any $C > 0$ and $\delta \in (0,1)$, there exists $\varepsilon = \varepsilon(C, \delta) > 0$ and $c = c(C, \delta) > 0$ such that every C-Ramsey graph G on n-vertex contains an (ε, δ)-rich subgraph on $\Omega(n)$ vertices.*

Proof. Let $K = K(C)$ be a large constant to be determined. Let $c \leq (\delta/4)^K$, and $\varepsilon = 1/K$. Suppose for contradiction that any vertex subset of size at least cn fails to induce an (ε, δ)-rich subgraph. We will inductively construct a sequence of disjoint vertex sets $S_1, \ldots, S_K$ of size $(cn)^\delta$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 4\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 4\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \subset G_1 \subset \cdots \subset G_K$, such that $|V(G_i)| \geq (\delta/4)|V(G_{i-1})|$ and $S_i \subseteq V(G_{i-1})$ for all i.

We first show that this construction suffices. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size $(cn)^\delta$ and $\varepsilon = 1/K$,

$$d(S) \leq \frac{\binom{(cn)^\delta}{2}}{\binom{r(cn)^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 4\varepsilon \leq \frac{6}{K}.$$

Note that $|S| \geq (cn)^\delta$, and so Theorem 1.3 yields a constant $\gamma > 0$ and a homogeneous set A in S of size

$$|A| \geq \frac{\gamma K}{6\log(K/6)} \log|S| \geq \frac{\gamma\delta K}{6\log(K/6)} \log cn = \frac{\gamma\delta K}{6\log(K/6)} \log n + o(\log n),$$

as $n \to \infty$. But then $\frac{\gamma\delta K}{6\log(K/6)} > C$ for sufficiently large K, and so $|A| > C\log n$, contradiction.

We now construct the sets S_i and G_i inductively. Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. Since $|V(G_{i-1})| \geq (\delta/4)^{i-1}n \geq cn$ and G_{i-1} is not (ε, δ)-rich by assumption, there exist $W_i, Y_i \subseteq V(G_{i-1})$ with $|W_i| \geq \delta|V(G_{i-1})|$ and $|Y_i| \geq (cn)^\delta$, such that for each $y \in Y_i$ we have $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ or $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon|W_i|$. Assume without loss of generality that $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ for half the vertices $y \in Y_i$, and let S_i be the corresponding subset of Y_i. Put $T_i = W_i \setminus S_i$ and note that $|T_i| \geq |W_i|/2$. Partition T_i into $V_i \sqcup Z_i$ where $V_i = \{v \in T_i : d(v, S_i) \leq 4\varepsilon\}$, and let $G[V_i]$. We need to show that $|V_i| \geq (\delta/4)|V(G_{i-1})|$. Observe that for all $y \in S_i$,

$$d(y, T_i) = \frac{d_{T_i}(y)}{|T_i|} < \frac{\varepsilon|W_i|}{|W_i|/2} = 2\varepsilon.$$

But then

$$4\varepsilon|Z_i| < \sum_{v\in Z_i} d(v,S_i) \le \frac{e(T_i,S_i)}{|S_i|} = \frac{|T_i|}{|S_i|}\sum_{y\in S_i} d(y,T_i) \le 2\varepsilon|T_i|,$$

so $|Z_i| < |T_i|/2$. It now follows that $|V_i| \ge |T_i|/2 \ge |W_i|/4 \ge (\delta/4)|V(G_{i-1})|$. Since $d(S_i,U) < 4\varepsilon$ for all $U \subseteq V_i$, condition (i) holds for S_i. In the case that $|\overline{N_{G_{i-1}}(x_1)} \cap W_i| < \varepsilon|W_i|$, set $V_i = \{v \in T_i : d(v,S_i) > 1-4\varepsilon\}$, which yields $|V_i| \ge (\delta/4)|G_{i-1}|$ and S_i satisfying condition (ii). This completes the induction. □

As mentioned in the start of the chapter, we fill find many induced subgraphs $G[U]$ whose edge counts are well-separated from each other. More specifically, Theorem 3.1 will be an immediate consequence of the following lemma, which we will dedicate the rest of this chapter to prove.

Lemma 3.4. *Let $C > 0$. Then there exists $c = c(C) > 0$, such that for any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \le m \le 2cn^2$, there are disjoint subsets $U, W \subseteq V(G)$ with $|e(U) - m| = O(n^{3/2})$ and $|W| = O(\sqrt{n})$ such that*

$$|\{e(U \cup Z) : Z \subseteq W\}| = \Omega(n^{3/2}).$$

We first quickly show how this implies Theorem 3.1.

Proof of Theorem 3.1. Note that for any U, W given by Lemma 3.4,

$$|e(U \cup W) - e(U)| = O(n^{3/2}),$$

as $|W| = O(\sqrt{n})$ and each $w \in W$ has at most n neighbours in $U \cup W$. But then the $\Omega(n^{3/2})$ values of $e(U \cup Z)$ for $Z \subseteq W$ are contained in an interval of length $O(n^{3/2})$ centered at $e(U)$. Partition the interval between cn^2 and $2cn^2$ into $\frac{c}{k}\sqrt{n}$ subintervals of length $kn^{3/2}$, for some large enough constant k to be determined. Now apply Lemma 3.4 to a suitable choice of m in each subinterval so that the obtained values of $e(U \cup Z)$ for $Z \subseteq W$ lie in the same subinterval as m, which is allowed for sufficiently large k. Since we applied Lemma 3.4 to $\Omega(\sqrt{n})$ disjoint subintervals, we obtain $\Omega(n^2)$ distinct sizes of induced subgraphs. □

4 The Erdős–McKay Conjecture

5 Concluding Remarks

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A Supplementary Proofs